Paper B : REACT Protocol and Feasibility: Analysis Plan

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1 Primary Proximal Effect Estimand

The IRB protocol defines the primary estimand as follows (Section 15.0 or §15.0, Primary analysis):

> "The primary estimand is the causal excursion effect: the average difference in the proximal outcome in the window following a decision point at which a prompt was delivered versus not delivered, averaged over available decision points."

Formally, let t index decision points for participant i during the micro-randomization period (study weeks two through five; §9.3). Let $I_{it} \in \{0, 1\}$ indicate decision point availability, and $A_{it} \in \{0, 1\}$ denote randomized assignment to prompt delivery. Let $Y_{i,t+1}$ be the proximal outcome measured in the time-locked outcome window (§2), and H_{it} denote the participant's history accrued up to the decision point. The causal excursion effect, marginal over history, is defined as:

$$\beta_0 = \mathbb{E}\left[Y_{i,t+1}(\bar{A}_{i,t-1}, 1) - Y_{i,t+1}(\bar{A}_{i,t-1}, 0) \;\middle|\; I_{it} = 1\right], \tag{1}$$

where $\bar{A}_{i,t-1}$ represents the participant's prior treatment history, ensuring the expectation averages across the causal histories generated by the trial itself [1]. It captures an excursion from the protocol at a single decision point rather than the cumulative policy effect of sustained prompting. When evaluated with a moderator vector $S_{it} \subseteq H_{it}$, the moderated causal excursion effect $\beta(S_{it})$ conditions on S_{it} (§5).

Treatment is the randomized assignment. In the analysis pipeline,

$$A_{it} = \mathbf{1}\{\texttt{randomization_draw} < \texttt{randomization_probability}\}, \tag{2}$$

computed from the values required to be logged at every decision point (§9.3: "Every decision point, whether or not a prompt is sent, is logged with its randomization probability, the draw outcome, the eligible message pool, the selected message, and delivery timestamps").

DISCREPANCY: Assignment vs. Delivery

The protocol's estimand contrasts decision points "at which a prompt was delivered versus not delivered." The software architecture separates these states:

1. `JITAILog.send_prompt` is reset to `False` following a successful draw when no message can be selected (`backend/app/tasks.py::_evaluate_user`, empty-pool branch).
2. A failed push keeps `send_prompt = True` with `status = 'failed'` (`notification_service.mark_delivery_failed`).

Contrasting delivered prompts conditions on post-randomization events, violating exchangeability and breaking the statistical guarantees of randomization.

2 Proximal Outcome Window

2.1 Time-Locked Outcome Assessment

Protocol §9.2:

> "During the micro-randomization period, the first check-in within two hours after any decision point presents the three behavior domains together for a time-locked outcome window; if no scheduled check-in falls in that window the app inserts one brief extra check-in, capped at four per day."

Protocol §10.0 (Schedule of Measures):

> "During the micro-randomization period, the first check-in within two hours after any decision point presents all four domain item sets together for a time-locked outcome window."

As Built. Every available decision point opens a time-locked outcome window at `decision_made_at`, regardless of assignment:

Outcome check-ins share the protocol daily cap of four. The 60-minute delay or prompt cool down guaranties that the measurement timing remains independent of the assignment arm. Without this delay, prompts direct participants to open the application, systematically inducing immediate responses in the active arm while control observations wait for the next scheduled prompt.

Analysis Definition. For each available decision point, the proximal outcome is derived from the first completed check-in linked to it (`source_jitai_log_id`, type `post_prompt` or `extra_check_in`) satisfying:

$$\texttt{decision_made_at} + 60 \text{ min} \leq \texttt{sent_at} \leq \texttt{decision_made_at} + 2 \text{ h}. \tag{3}$$

Values are extracted from `EMAItemResponse.value_numeric` / `value_choice` indexed by `sub_item_id`. Aggregate EMA-level mood, stress, and energy fields are excluded, as the submission handler does not populate them. Decision points lacking a completed check-in within this interval are treated as missing (§6).

Post-Decision Time	Content Served by `/ema/next/`	Notification Behavior
0 to 60 min	Suppressed, except the C_0 post-prompt rating after a delivered prompt (§10.0). Standard scheduled check-ins are held back (`reason: outcome_window_pending`) in both arms.	None; scheduled check-in notifications are actively suppressed in both arms.
60 min to 2 h	Outcome check-in: current emotions/situation alongside four domain item sets (`POST_PROMPT_ITEM_IDS`), linked to the decision point (`source_jitai_log`). Served as `post_prompt` or `extra_check_in`.	At +60 min (if uncompleted): one visible reminder sent identically to both arms (`_maybe_send_outcome_reminder`).
$>$ 2 h (or completed)	Ordinary scheduled check-ins resume.	Standard check-in reminders resume.

Table 1: Outcome window lifecycle and notification dispatch rules.

DISCREPANCY: Domain Count Discrepancy

Section 9.2 cites "the three behavior domains," whereas §10.0 specifies "all four domain item sets." The deployed build serves `POST_PROMPT_ITEM_IDS = ['B1', 'B2', 'B4', 'B5', 'B6', 'B7']`, covering the four rotating domains (B4 eating, B5 emotion regulation, B6 alcohol, B7 online activity) plus valence and contextual situation (`backend/app/ema_catalog.py`).

2.2 Primary Behavioral Outcomes

Protocol §15.0:

> "Proximal outcomes are drawn from the time-locked check-in within two hours after the decision point: self-reported urge to post reactively, self-reported reactive online behavior, self-reported drinking and drinking urge, and self-reported emotional or uncontrolled eating, each analyzed separately."

Outcomes are fitted within individual WCLS models. Binary outcomes are estimated on the risk difference (RD) scale; log relative-risk models [5] serve as secondary specifications.

3 Weighted-and-Centered Least Squares Analysis

Protocol §15.0:

> "This is estimated by weighted and centered least squares (Boruvka et al., 2018), with weights derived from the known randomization probability of 0.5 and centering on the randomization probability, which yields a consistent estimate of the marginal excursion effect without requiring a model for the outcome under no treatment. Standard errors use a sandwich estimator with small-sample correction, clustering at the participant level."

3.1 Treatment Assignment and Weighting

Randomization follows a two-stage sequential draw (`backend/app/tasks.py::_evaluate_user`):

Construct	Candidate Sub-Item(s)	Scale	Recall Period	Model
Urge to post	`B7_urge`	Likert 1–7	Now	Linear WCLS
Online behavior	`B7_posted`; `B7_charged`; `B7_self_stopped`	Binary; Likert 1–7; 3-level choice	"Past few hours"	Binary WCLS (RD)
Drinking	`B6_consumed`; `B6_drink_count`	Yes/No; Count	"Today"	See note
Drinking urge	`B6_urge`	Likert 1–7	Now	Linear WCLS
Emotional eating	`B4_emotion_influence`; `B4_loss_of_control`	Likert 1–7 (if `B4_ate` = Yes)	"Since last check-in"	See note

Table 2: Operationalization candidates for primary behavioral outcomes.

1. **Prompt assignment:** Draw $U_{it} \sim \mathcal{U}(0,1)$; assign $A_{it} = \mathbf{1}\{U_{it} < p_{it}\}$, where $p_{it} =$ `JITAI_RANDOMIZATION_PROBABILITY` $= 0.5$.
2. **Message arm:** If $A_{it} = 1$, a secondary draw allocates coping intervention versus active control ($p_{\text{arm}} = 0.5$).

Let $\tilde{p}$ denote the sample mean of logged probabilities p_{it}. Each decision point is weighted by:

$$W_{it} = \left(\frac{\tilde{p}}{p_{it}}\right)^{A_{it}} \left(\frac{1-\tilde{p}}{1-p_{it}}\right)^{1-A_{it}}. \tag{4}$$

When $p_{it} \equiv 0.5$, $W_{it} = 1$ for all observations. Retaining per-row evaluation ensures stability should the environment parameter vary during deployment. Positivity ($0 < p_{it} < 1$) is verified prior to model fitting.

3.2 Centered Treatment Indicator

Treatment enters the model centered as $(A_{it} - \tilde{p})$. For any history H_{it} and availability $I_{it} = 1$:

$$\mathbb{E}\left[W_{it}(A_{it} - \tilde{p}) \mid H_{it}, I_{it} = 1\right] = p_{it}\frac{\tilde{p}}{p_{it}}(1-\tilde{p}) + (1-p_{it})\frac{1-\tilde{p}}{1-p_{it}}(0-\tilde{p}) = 0. \tag{5}$$

Consequently, the weighted centered indicator is orthogonal to arbitrary functions of baseline covariates. Specification errors in the control terms $g(H_{it})$ affect precision rather than asymptotic consistency, assuming outcome observation remains independent of assignment.

3.3 Estimation of the Causal Excursion Effect

The parameter vector $\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\alpha}}^\top, \hat{\boldsymbol{\beta}}^\top)^\top$ solves the weighted estimating equations across available decision points with observed outcomes:

$$\sum_{i=1}^{n} \sum_{t} I_{it} W_{it} \left(Y_{i,t+1} - g(H_{it})^\top \boldsymbol{\alpha} - (A_{it} - \tilde{p}) f(S_{it})^\top \boldsymbol{\beta} \right) \begin{pmatrix} g(H_{it}) \\ (A_{it} - \tilde{p}) f(S_{it}) \end{pmatrix} = \mathbf{0}. \tag{6}$$

- **Primary marginal model:** $f(S_{it}) = 1$, setting $\boldsymbol{\beta} = \beta_0$.
- **Control vectors $g(H_{it})$:** Intercept, trigger-level affect valence (`B1_valence`), stress (`B2_stress`), hour of day (Eastern, centered at 15:00), and study day (centered at mid-study).
- **Working independence:** Observations within participant are assumed independent in the point estimating equations to preserve consistency without requiring correct time-series covariance specifications; repeated measures clustering is addressed in the variance estimator.

3.4 Statistical Inference and Power

Covariance Estimation. Inference uses the robust sandwich estimator $\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}) = B^{-1} M B^{-1}$ with the Mancl–DeRouen small-sample correction [3]:

$$B = \sum_{i=1}^{n} X_i^\top W_i X_i, \qquad M = \sum_{i=1}^{n} X_i^\top W_i (I - H_{ii})^{-1} \boldsymbol{r}_i \boldsymbol{r}_i^\top (I - H_{ii})^{-\top} W_i X_i, \tag{7}$$

where $H_{ii} = X_i B^{-1} X_i^\top W_i$. Test statistics are referred to Student's t distribution with $n - \dim(\boldsymbol{\alpha}) - \dim(\boldsymbol{\beta})$ degrees of freedom.

Sample Size and Availability Realignment. The protocol assumed an availability of 0.6 (§7.0). However, the deployed architecture requires MSSD to exceed the participant's expanding 80th percentile, fixing true availability at approximately 0.20. Following [2], the non-centrality parameter for the primary hypothesis test is:

$$\lambda = N d^2 p(1-p) \sum_{t} \mathbb{E}[I_t], \qquad \text{Power} = \Pr\left(F_{1,N-q-1}(\lambda) > F^{\text{crit}}_{1,N-q-1} \right). \tag{8}$$

Protocol §7.0 (Sample Size):

> "Sample size follows the approach for micro-randomized trials in Liao et al. (2016). With about 80 participants, micro-randomization active for roughly four weeks, up to six decision points per day, an assumed availability near 0.6, and a send probability of 0.5, each participant contributes on the order of a hundred available randomized decision points, giving several thousand across the sample and about half that with a prompt delivered."

The protocol's stated power claim still holds at the empirical availability of 0.20. Protocol §7.0 specifies detection of "a standardized proximal main effect in the range of 0.10 to 0.15 at about 80 percent power." With $N = 80$, 28 days of micro-randomization (six decision points daily), and $p = 0.50$, the primary test has 80% power (two-sided $\alpha = 0.05$) to detect standardized effects of:

Scenario ($N = 80$)	**Avail.**	**DP/Person**	**Total DP**	**MDE** (80% **Power**)	**Power at** $d = 0.10/0.15$
Protocol target (Avail. = 0.6)	100.8	8,064	0.063	0.99 / 1.00	
Deployed system (Avail. = 0.2)	33.6	2,688	0.109	0.73 / 0.97	
+75% completion benchmark	25.2	2,016	0.126	0.60 / 0.91	
+75% outcome response rate	18.9	1,512	0.146	0.48 / 0.82	
Same, with $N = 70$ retained	18.9	1,323	0.156	0.43 / 0.77	

Table 3: Statistical power across study engagement scenarios.

- 0.109 under full engagement;
- 0.126 at the 75% check-in completion benchmark;
- 0.146 when 75% of outcome check-ins are answered;
- 0.156 if sample retention falls to $N = 70$.

These values fall within the upper end of the protocol's target interval (0.10–0.15). In synthetic evaluations (1,000 simulated cohorts), empirical power was 0.784–0.798 across benchmark scenarios with nominal Type I error rates of 0.042–0.058.

DISCREPANCY: Protocol Sample Estimates

The protocol estimates "on the order of a hundred available randomized decision points" per participant, yielding "about two to three prompts per day." Under an availability of 0.20, empirical yield is $\approx$ 19–34 available decision points per participant, corresponding to $6 \times 0.2 \times 0.5 \approx 0.6$ prompts daily.

4 Availability and Decision Points

Protocol §9.3:

> "Availability. A decision point is unavailable, and no prompt is delivered, when the participant is asleep according to device data, when the device has not synced recently enough to evaluate the signal, when a prompt has been delivered within the preceding 60 minutes, when the daily cap has been reached, or when a safety override is in effect. Unavailable decision points are recorded and are handled explicitly in the analysis rather than being ignored."

A decision point is instantiated by any completed EMA containing valence, stress, and arousal. The point is flagged available ($I_{it} = 1$) only when:

1. The MSSD trigger threshold is exceeded ($>$ 80th percentile on ≥ 3 prior values);
2. The 60-minute system cooldown has elapsed;

3. Daily delivered prompt count is below four;
4. Study day ≥ 7 (completion of run-in phase);
5. No active distress flag is present.

DISCREPANCY: System Availability Implementations

- **Sleep/sync status:** Wearable pipelines remain unintegrated in triggering; participants are wake-confirmed via EMA completion.
- **Cooldown scope:** The deployed engine tracks cooldown across all eligible points rather than delivered prompts alone.
- **Decision point frequency:** The protocol cites "up to six decision points per day." In production, post-prompt and extra check-ins can themselves trigger decision points, yielding up to ten potential opportunities daily.

4.1 Safety Override and Distress Flags

Protocol §11.0:

> "During the intensive period, any distress signal suspends prompt randomization for that participant and the app shows a resource card instead of a coping message."

Flag Level	Trigger Condition	Duration	System Response
Momentary	Scale boundary on EMA (`B1_valence = 1`, `B2_stress = 7`, `B1_affect_sad = 5`, `B1_affect_anxious = 5`)	24 hours	Randomization suspended; local resource card displayed.
Baseline	Scored Qualtrics screen (PHQ-9 self-harm, AUDIT-C, SCOFF, PGSI)	Until contact logged	Randomization suspended; study team alerted via Qualtrics.

Table 4: Safety override classifications and intervention routing.

5 Moderation Analyses

5.1 Model Specification

Moderated causal excursion effects are estimated by specifying $f(S_{it}) = (1, S_{it})^\top$. Covariate set $g(H_{it})$ must strictly include all main effects of S_{it}; omitting S_{it} from $g(H_{it})$ causes confounding leakage into $\hat{\boldsymbol{\beta}}$ when missingness depends on assignment. Continuous moderators are centered.

5.2 Confirmatory Tier 1 Moderators

DISCREPANCY: DERS-16 Classification

DERS-16 is specified as a Tier 1 confirmatory moderator in protocol §1.0 but is omitted from §15.0.

Tier 1 Moderator Scale	Paired Proximal Behavioral Outcome
SUPPS-P Negative/Positive Urgency	Urge to post reactively; reactive online behavior
DMQ-R Coping Drinking Motives	Alcohol consumption; drinking urge
TFEQ-R18 Emotional/Uncontrolled Eating	Emotional or uncontrolled eating
DERS-16 Total Score	Evaluated across all four domain outcomes
MAIA-2 (Noticing, Body Listening)	Evaluated across all four domain outcomes

Table 5: Pre-specified Tier 1 moderation pairings.

5.3 Message-Type Sub-Studies

Message comparative effectiveness is analyzed in two tiers:

1. **Coping vs. Active Control (Primary):** Evaluated via WCLS restricted to $A_{it} = 1$ using `arm_randomization_probability`.
2. **Technique Families (CBT vs. ACT):** Evaluated within category pools conditioning on the empirical assignment probability $P(\text{CBT} \mid \text{pool}) = \frac{\#\text{CBT}}{\text{Total Pool}}$.

6 Missing Data and Sensitivity Analyses

Missingness operates at two tiers:

- **Unobserved decision points:** Handled by definition within the estimand conditioning on $I_{it} = 1$.
- **Missing outcome responses:** Available decision points lacking a valid follow-up EMA within [+60 min, +2 h] are excluded in complete-case estimation.

Label	Sensitivity Dimension	Analytical Execution	Design Boundary
S1	Outcome Window	Restrict outcome follow-up to +90 min post-trigger.	Cannot evaluate $<$ 60 min due to design delay.
S2	Availability Definition	(a) Exclude extra check-in triggers; (b) base cooldown solely on delivered prompts.	Cannot expand to untriggered points.
S3	MSSD Threshold	Re-evaluate threshold at 85th and 90th quantiles.	Cannot back-evaluate $<$ 80th percentile.
S4	Compliance	Exclude participants with $< 75\%$ EMA completion.	Evaluated as a subcohort model.
S5	Outcome Missingness	Inverse-probability weighting conditional on A_{it}, time, and trigger affect.	Sensitivity check if dropout differs by arm.
S6	Legacy Protocol	Evaluate pre-cutover observations via scheduled EMAs.	Disaggregated entirely from main trial.

Table 6: Suite of pre-specified sensitivity analyses.

References

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